

CATERPILLAR® MACHINE DIAGNOSTICS

INDUSTRIAL INSPECTION & SYSTEM CONFIGURATION AUDIT

Machine ID:	CAT-HEX-320	Date of Inspection:	2026-07-20
Machine Name:	CAT Hydraulic Excavator	Lead Engineer:	Sarah Engineer
Manufacturer:	Caterpillar Inc.	Model / Category:	320-GC / CAT Hydraulic Excavator
Diagnostic Status:	Warning (Score: 75%)	Engine Version:	v2.4-deterministic

System Hardware & Firmware Configuration Audit

Below is a side-by-side comparison of the active telemetry and local software modules against original factory reference parameters.

Parameter	Factory Reference	Current Machine	Audit Status
Firmware Version	v4.2.1-lts	v4.2.1-lts	MATCHED
PLC Version	v3.12-revB	v3.10-revA	MISMATCH (WARNING)
CPU Architecture	Intel Atom E3950	Intel Atom E3950	MATCHED
RAM (Memory)	8GB DDR3	4GB DDR3	MISMATCH (WARNING)
Storage	64GB SSD	64GB SSD	MATCHED
Sensor Count	12	12	MATCHED
Communication Ports	USB, COM1, COM2, Ethernet	USB, COM1, COM2, Ethernet	MATCHED
Installed Modules	Analog Input, Digital IO, CAN Bus controller	Analog Input, Digital IO, CAN Bus controller	MATCHED
Operating Temperature	Under 70 C	52.4 C	MATCHED
Power Supply Status	Stable	Stable	MATCHED

Detected Mismatches & Anomalies

1. PLC Version Mismatch [WARNING]

PLC configuration version mismatch. Expected 'v3.12-revB', found 'v3.10-revA'.

2. RAM (Memory) Mismatch [WARNING]

System memory allocation mismatch. Expected '8GB DDR3', active size is '4GB DDR3'.

AI-Powered Diagnostics Analysis

Machine Safety & Health Assessment

DEGRADED / ACTION REQUIRED (Health: 75%): Mismatches or anomalous sensor readings detected. Operational risks present.

Root Cause Analysis

The diagnostic engine flagged the following configuration mismatches: - PLC Version: PLC configuration version mismatch. Expected 'v3.12-revB', found 'v3.10-revA'. - RAM (Memory): System memory allocation mismatch. Expected '8GB DDR3', active size is '4GB DDR3'. Typical root causes include: field component replacement with unauthorized modules, out-of-date PLC programming flashes, firmware regression during maintenance cycles, or thermal cooling degradation.

Suggested Maintenance Worksteps

1. Connect Service Tool (CAT ET or Siemens TIA Portal) and flash target firmware and PLC image profiles matching reference configurations.
2. Audit active hardware modules. Replace generic CPU boards, RAM modules, or expansion cards with OEM-authorized parts.

Crucial Safety Notes

- Verify power source disconnect before opening PLC enclosures.
- Ensure constant power backup is connected to the PLC logic controllers during the programming and flashing sequence.
- Follow Electrostatic Discharge (ESD) wrist strap procedures when handling exposed logic boards or memory chips.

Verification & Troubleshooting Steps

1. Run firmware CRC checksum validation command. Check serial console output for boot loader warnings.
2. Perform full system diagnostic self-test from the service shell. Verify device tree registration of all local modules.

PREPARED BY:	REVIEWED & SIGNED BY:
Automated Diagnostic Engine CAT Diagnostic Systems	Lead Engineer: Sarah Engineer CAT Certified Field Inspector